

# ARIA INSTITUTIONAL RISK REPORT

Advanced Retail Intelligence & Analytics — v5

Generated: April 01, 2026 at 11:45 UTC

## EXECUTIVE SUMMARY

| Metric               | Value     | Status          |
|----------------------|-----------|-----------------|
| Crisis Level         | CRITICAL  | ■ CRITICAL      |
| RED Alerts           | 8         | 8/15 pairs      |
| Portfolio Protection | 50.0%     | Guardian Active |
| Worst GAN Scenario   | -17.65%   | Stress Tested   |
| Total Hedge Cost     | \$5000.00 | Auto-sized      |

## EXTREME VALUE THEORY — TAIL RISK (POT/GPD Method)

Using Peaks-Over-Threshold with Generalized Pareto Distribution. VaR99 = worst expected loss on 1 in 100 trading days. Expected Shortfall = average loss beyond VaR99.

| Asset | Threshold | VaR 99% | Exp. Shortfall | Shape ( $\xi$ ) | Risk Level |
|-------|-----------|---------|----------------|-----------------|------------|
| ETH   | 4.18%     | 9.59%   | 11.83%         | -0.0215         | EXTREME    |
| TSLA  | 3.96%     | 9.18%   | 12.05%         | 0.1083          | EXTREME    |
| NVDA  | 3.33%     | 8.34%   | 10.91%         | 0.0779          | HIGH       |
| BTC   | 2.86%     | 6.33%   | 8.06%          | 0.0613          | HIGH       |
| AAPL  | 1.83%     | 4.77%   | 6.26%          | 0.0678          | MODERATE   |
| GLD   | 1.39%     | 4.16%   | 5.99%          | 0.1890          | MODERATE   |

## TAIL DEPENDENCE — CRISIS CORRELATIONS

Tail dependence measures how assets co-move during extreme events. Crisis correlation shows actual correlation during worst 20% market days. Higher tail dependence = more dangerous to hold both assets simultaneously.

| Pair      | Normal Corr | Crisis Corr | Tail Dep Coeff | Diversification |
|-----------|-------------|-------------|----------------|-----------------|
| AAPL/GLD  | +0.025      | +0.041      | 0.140          | GOOD            |
| AAPL/NVDA | +0.378      | +0.303      | 0.319          | MODERATE        |
| AAPL/TSLA | +0.438      | +0.389      | 0.379          | MODERATE        |
| BTC/AAPL  | -0.038      | -0.012      | 0.080          | GOOD            |
| BTC/ETH   | +0.820      | +0.717      | 0.699          | POOR            |
| BTC/GLD   | +0.044      | +0.042      | 0.040          | GOOD            |
| BTC/NVDA  | -0.106      | -0.129      | 0.040          | GOOD            |
| BTC/TSLA  | -0.002      | -0.029      | 0.120          | GOOD            |
| ETH/AAPL  | -0.033      | -0.063      | 0.060          | GOOD            |
| ETH/GLD   | +0.003      | +0.011      | 0.020          | GOOD            |

## GAN STRESS SCENARIOS — TOP 5 WORST CASES

10,000 synthetic crisis scenarios generated by Generative Adversarial Network trained on historical crisis periods. No Tanh constraint — true outlier generation enabled.

| Scenario | Max Drawdown | BTC | ETH | AAPL | NVDA | TSLA | GLD |
|----------|--------------|-----|-----|------|------|------|-----|
|----------|--------------|-----|-----|------|------|------|-----|

|            |         |        |        |       |        |       |       |
|------------|---------|--------|--------|-------|--------|-------|-------|
| Scenario 1 | -16.13% | -12.3% | -16.1% | -0.1% | 8.2%   | 3.5%  | 0.6%  |
| Scenario 2 | -16.13% | -11.5% | -16.1% | 0.0%  | 7.6%   | 4.7%  | -0.3% |
| Scenario 3 | -13.28% | -1.1%  | -1.3%  | -1.3% | -13.3% | -1.2% | 3.8%  |
| Scenario 4 | -13.24% | -9.7%  | -13.2% | -0.4% | 4.3%   | 2.4%  | 0.0%  |
| Scenario 5 | -12.70% | -9.1%  | -12.7% | -1.8% | 5.4%   | -0.9% | 0.5%  |

### LAYER 1 — RISK PARITY POSITION SIZING

Position sizes calculated using inverse Expected Shortfall (Risk Parity). Assets with higher tail risk receive proportionally smaller allocations. Formula:  $\text{Position} \propto 1/\text{ES} \times \text{Regime\_Multiplier} \times \text{Kelly\_Fraction}$

| Asset | Position Size | Weight | Regime   |
|-------|---------------|--------|----------|
| GLD   | \$3525.87     | 0.235  | SIDEWAYS |
| AAPL  | \$3376.49     | 0.225  | SIDEWAYS |
| BTC   | \$2622.49     | 0.175  | SIDEWAYS |
| NVDA  | \$1935.72     | 0.129  | SIDEWAYS |
| ETH   | \$1786.46     | 0.119  | SIDEWAYS |
| TSLA  | \$1752.97     | 0.117  | SIDEWAYS |

### LAYER 3 — GUARDIAN AGENT PROTECTION ANALYSIS

| Metric                 | Value     |
|------------------------|-----------|
| Crisis Level           | CRITICAL  |
| RED Alerts             | 8         |
| Worst Scenario         | -17.65%   |
| Loss Without Hedge     | \$1764.55 |
| Guardian Hedge Profit  | \$882.27  |
| Net Loss With Guardian | \$882.27  |
| Protection Coverage    | 50.0%     |
| Total Hedge Cost       | \$5000.00 |